SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

SELADEVS UTHM

Project / Solution Name

SHIFTATLAS 3D

Name (Leader)

MUHAMMAD HAZIQ FARHAN BIN NORDIN

Email & Phone (Leader)

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University Affiliation(s)

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Live Prototype URL

https://sdc-2026-seladevsuthm.replit.app/
YouTube: https://youtu.be/e8jtdJy6deo

Code Repository URL (public / access granted to judges)

https://github.com/hzqfarhan/SDC2026-SELADEVSUTHM

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

We turn DOSM CSV/GeoJSON, institution facts, Wikidata coordinates and arXiv metadata into dated, hash-traceable snapshots. Our Next.js/React application runs local TF-IDF, keyword review, weighted skill coverage, prerequisite learning and uncertain scenarios; React Three Fiber renders the connected 3D worlds alongside text views. We built a Node server for the static export and optional semantic/Council gateways. Fictional-text consent and a demo access code gate cloud requests; we keep API secrets and trusted catalogues server-side. Our RunPod workers use pinned multilingual-e5-base for relevance and Qwen3-4B-Instruct-2507 for four sequential reviews. We validate schemas, known identifiers and exact profile quotations, then restore learning prerequisites. All Council viewpoints use one model, so the critic is not independent. We preserve proficiency levels and rubric scores. Profiles and activity stay locally stored, and uploaded files stay in the browser. We use bounded memory caches, request limits and timeouts, without treating them as billing caps. We verified an E5 image build and synthetic CUDA jobs for E5 and Qwen profile review. Our public Next pages and both production API status routes return HTTP 200. The E5 gateway completed a six-role CUDA request. Our local flow works independently.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Next.js 16.4.2 React/DOM 19.3.1	Static application; UI and state	MIT
Three.js 0.180.0 R3F 9.8.2 Drei 10.7.9	Procedural 3D rendering, camera and labels	MIT
TypeScript 5.9.3 tsx 4.23.15	Type checks; test/script runner	Apache-2.0; MIT
PDF.js 6.4.299	Local PDF text extraction	Apache-2.0
Mammoth 1.13.0	Local DOCX raw-text extraction	BSD-2-Clause
Lucide 0.468.0	Interface icons	ISC

We used Node/npm, ReportLab/pypdf, Poppler, Artifact Tool, FFmpeg and Windows speech for our build and submission. Notices are in our repository.

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless API	We completed E5 and Qwen-profile CUDA jobs; full-flow acceptance is separate	Sponsor credit only; direct CUDA fixtures
OpenDOSM public endpoints	We fetch CSV/GeoJSON for dated browser snapshots	Free public data; no key
arXiv API	We snapshot dated metadata; titles may enter consented cloud requests	Free public endpoint
Wikidata / MQA public sources	We map coordinate facts and institution-state tags with separate coverage	CC0 / MQA source terms
Google Fonts	DM Sans / Manrope with system fallbacks	Public font service
OpenAI runtime API / analytics	We use no OpenAI runtime API; development tools are declared separately	Not applicable
TalentCorp / ILO / O*NET / learning providers	Public source links; trusted catalogue references	Public linked material

We exclude private credentials and retain public endpoint references and hashes with our snapshots.

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	We published our interface; public landing and atlas routes return HTTP 200	Autoscale: 2 vCPU / 4 GiB RAM; Asia
GitHub	We publish our source, history, notices and fallback build	Repository link in Section 1
Node/static export	We provide UI, trusted catalogues, access checks and optional APIs	Public API status routes respond; E5 gateway completed on CUDA
OpenAI Sites workflow	We tested packaging as a development workflow	Our published application uses Replit
RunPod Serverless	We completed E5 and Qwen-profile CUDA fixtures	Sponsor credit only; full-flow acceptance is separate
Database / account services	We use local state and bounded server-memory caches	Not applicable
Cloud status	We retain public E5 gateway and direct Qwen-profile CUDA receipts	Both public APIs respond; complete Council acceptance is separate

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
multilingual-e5-base	intfloat / Hugging Face	We built the pinned image and completed a six-role public CUDA request.	MIT
Qwen3-4B-Instruct-2507	Qwen / Hugging Face	We completed profile-stage CUDA inference (1/4); full-flow acceptance is separate.	Apache-2.0
Local TF-IDF / cosine, v1	Authored six-role corpus	We fit classical retrieval locally; it does not assess proficiency	Project Apache-2.0
Weighted rubric / title-topic rules	Authored targets/weights; O*NET/arXiv refs	We explain deterministic coverage; optional weight adjustment is capped at 25%	Project / source terms
Scenario coefficients, v1	Authored hypotheses; ILO context	We authored hypotheses without training a forecast	Project Apache-2.0
Council validation	Project gateway / worker code	We validate known IDs, exact quotes and schemas, then restore prerequisites	Project Apache-2.0
Execution boundary	No model training/fine-tuning	Our E5 and Qwen-profile fixtures establish runtime execution; model-quality evaluation is separate	Direct worker checks

Pinned E5 SHA: d128750597153bb5987e10b1c3493a34e5a4502a
Pinned Qwen SHA: cdbee75f17c01a7cc42f958dc650907174af0554

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
DOSM labour, 2026Q2	open.dosm.gov.my lfs_qtr_state	Public; aggregate	CC BY 4.0	N
DOSM state polygons	github.com/dosm-malaysia/data-open	Public; GeoJSON	Source terms; DOSM attribution	N
Wikidata coordinates	wikidata.org entity links in snapshot	Public structured facts	CC0	N
MQA directory facts	www2.mqa.gov.my/mqr	Public factual names/state tags	MQA source terms	N
arXiv metadata	export.arxiv.org/api/query	Public paper metadata	Author/feed terms; no full text	N
Authored role / SDG / learning catalogue	O*NET, UN and provider links in source register	Authored interpretations; references	Project Apache-2.0; reference terms	N
Synthetic profiles / resumes	public/data/synthetic-resume.txt	Synthetic; no real CVs	Original project material	N
17 official SDG icons	sdgs.un.org/goals	Public UN visual assets	UN usage guidance; not Apache	N

We use no private training data. Our source URLs, dates, geography and hashes are in docs/DATA_PROVENANCE.md and public/data.

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex / GPT-5.1 Sol	We used these tools for coding, research, testing, design and documents	Major
Qoder / Cortex Code (CoCo)	We used both tools for coding assistance	Supporting coding assistance
OpenAI image generation	We generated our app icon and low-poly illustrations; image-model version unavailable	Minor asset creation
Replit Agent; Windows speech / FFmpeg	We used setup scaffolding, restored our source, and assembled synthetic speech/video	Minor setup / video production

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We reviewed our core prototype and can explain its main components. We are completing our added-cloud review. We used OpenAI Codex extensively, with GPT-5.1 Sol, Qoder and Cortex Code (CoCo) for coding assistance. OpenAI image generation produced our icon and illustrations; Replit Agent assisted setup. Windows speech and FFmpeg produced our video. We completed synthetic CUDA jobs for E5 and Qwen profile review. Our tests cover schemas, quotations, references and failure paths. Our public Replit interface and both API status routes respond. Our E5 gateway completed a six-role CUDA request; model-quality evaluation is separate.

We passed 55 app tests, 18 offline worker tests, TypeScript and production build. Our public Next pages and both API status routes return HTTP 200. The public E5 gateway completed an uncached six-role CUDA request. Our direct Qwen profile fixture also completed on CUDA. Council uses a separate endpoint.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit / SDC	We published our interface and prepared the connecting Node/UI/API gateway	Public pages/APIs: HTTP 200; E5 gateway: CUDA
RunPod / SDC	We redeemed US$25 sponsor credit; E5 and Qwen-profile ran on CUDA	Execution / delay: E5 3.452 / 11.626 s; Qwen-profile 5.927 / 9.342 s
Other sponsor resources	We use no additional sponsor API or dataset	We use sponsor credit only and monitor our remaining balance
Deployment / endpoints	https://replit.com/@haziqfarhan174/SDC2026-SELADEVSUTHM E5: 7ib627oev3kwlb Council: yx2pzhigxtbnnm	E5 COMPLETED / CUDA; abdc818d-4447-4171-95d1-adafa18b47ce-e2

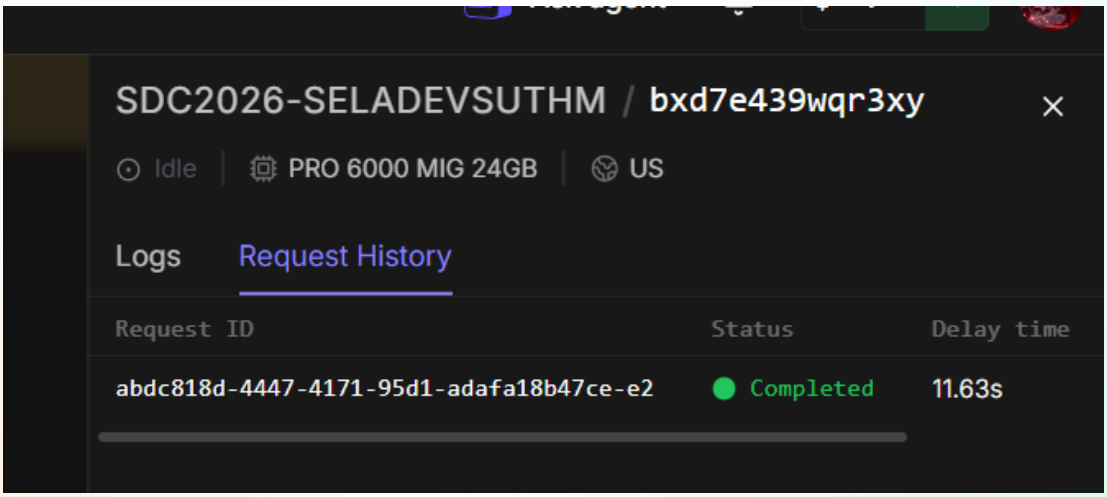
Our own-account Replit project and published Next interface are linked above. Both public API status routes returned HTTP 200; our E5 gateway completed an uncached six-role CUDA request.

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

We use Replit to publish our interface, trusted-catalogue Node gateway and streaming Council API. Our public E5 route completed a real CUDA request comparing a fictional learner with all six roles and dated research titles. RunPod also completed our direct Qwen profile-review fixture. We redeemed US$25 sponsor credit and disabled auto-pay. We retain the E5 image-build record, pinned model revisions and execution receipts. Replit keeps provider credentials server-side; our original rubric and learning planner remain local. We use sponsor credit only, bounded requests and a deployment recipe with zero minimum GPU workers.



Actual RunPod request history: COMPLETED; PRO 6000 MIG 24GB.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

We use synthetic profiles and fictional resumes. Resume files and extracted text stay in browser memory; only accepted skill levels and activity persist in localStorage. Optional cloud requests need separate fictional-data consent and a private demo code. The gateway sends selected fictional text and public role, course or research context to RunPod. Provider credentials remain server-side. It validates outputs before display and preserves the original skill rubric. Server caches have 60-minute validity and bounded entries; eviction and restart remove them. This does not certify provider-side deletion. We use no backend user database or product accounts. Replit publishing adds platform analytics. Our E5 application request used fictional text and public references. Competition contact details appear only in official submission documents.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Our ten skills and six O*NET roles define the prototype coverage. Malaysian transferability and multilingual relevance require evaluation. We author rubric weights and scenario coefficients, treating research titles as topic signals and E5 cosine similarity as relevance. Scenarios are hypothetical rather than calibrated forecasts. Our public E5 request and Qwen profile fixture establish execution; model quality and complete Council acceptance are separate checks. Qwen may hallucinate, and its four viewpoints share one model's weaknesses. Request limits and cancellation reduce usage without guaranteeing a monetary cap, so sponsor billing controls matter. Provider retention needs review. We distinguish state labour aggregates from institution coordinates and present SDG mappings as intended pathways. Pilot targets are evaluation goals, with assessed learning and employment outcomes for future studies. Human review and transparent coefficients remain central.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [ ] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [ ] All fields in this template have been completed within the word limits stated.

We reviewed our core, checked the functional fallback and reached our public Replit interface. E5 and Qwen-profile CUDA jobs completed. Public pages/APIs and the E5 gateway run; added-cloud review and full Council acceptance are separate.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

Signature

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Name: MUHAMMAD HAZIQ FARHAN BIN NORDIN

[Date]

25 October 2026